

4.0

Avaliação Parcial 1

Curso: Tecnologia em Telemática

Disciplina: Cálculo Diferencial e Integral 2

Professor: Sebastião Pontes Mascarenhas

Semestre: 2023.2

1.0

01. Determine a integral indefinida:

$$\int \frac{dx}{\sqrt[4]{x} - 1}$$

0.5

02. Determine a integral indefinida:

$$\int \frac{\sqrt{9 - x^2} \cdot dx}{x^2}$$

X

03. Determine a integral indefinida:

$$\int \frac{(4x - 7) \cdot dx}{x^2 - 5x + 6} ?$$

2.5

04. Calcule a integral indefinida:

$$\int x \cdot \cos x \cdot dx$$

$$1.) \int \frac{dx}{\sqrt[4]{x}-1} = \int \frac{4y^3 \cdot dy}{y-1} = 4 \int \frac{y^3}{y-1} \cdot dy =$$

TOME: $y = \sqrt[4]{x}$

Logo, $y^4 = x$

$\therefore \frac{dx}{dy} = 4y^3$

$\therefore dx = 4y^3 \cdot dy$

$$= 4 \int \left(y^2 + y + 1 + \frac{1}{y-1} \right) \cdot dy =$$

$$= 4 \left(\frac{y^3}{3} + \frac{y^2}{2} + y + \ln|y-1| \right) + K =$$

$$= 4 \left(\frac{\sqrt[4]{x}^3}{3} + \frac{\sqrt[4]{x}^2}{2} + \sqrt[4]{x} + \ln|\sqrt[4]{x}-1| \right) + K.$$

$D = d \cdot q + r$

$\frac{D}{d} = q + \frac{r}{d}$

$$\begin{array}{r} y^3 \overline{) 1y-1} \\ -y^3 + y^2 \\ \hline y^2 + y - 1 \end{array}$$

$$\frac{-y^2 + y}{1}$$

$$\frac{-y+1}{1}$$

$\therefore y^3 = (y-1) \cdot (y^2+y+1) + 1$

$\therefore \frac{y^3}{y-1} = (y^2+y+1) + \frac{1}{y-1}$

$$2.) \int \frac{\sqrt{9-x^2}}{x^2} \cdot dx = \int \frac{3 \cdot \cos \theta \cdot 3 \cdot \cos \theta \cdot d\theta}{9 \cdot \tan^2 \theta}$$

TOME $x = 3 \cdot \tan \theta$

$\frac{dx}{d\theta} = 3 \cdot \cos \theta$

$\therefore \sqrt{9-x^2} = \sqrt{9-9 \cdot \tan^2 \theta} =$

$= \sqrt{9(1-\tan^2 \theta)} = \sqrt{9 \cos^2 \theta} =$

$= 3 \cdot \cos \theta$

$$= \int \frac{9 \cdot \cos^2 \theta \cdot d\theta}{9 \cdot \tan^2 \theta} =$$

$$= \int \frac{(1 - \sin^2 \theta) \cdot d\theta}{\tan^2 \theta} =$$

$$= \int \left(\frac{1}{\tan^2 \theta} - \frac{\sin^2 \theta}{\tan^2 \theta} \right) \cdot d\theta =$$

$$= \int \frac{1}{\tan^2 \theta} \cdot d\theta$$

4. $\int x \cdot \cos x \cdot dx \Rightarrow \int \cos x \cdot x \cdot dx = \cancel{\sin x} \cdot x - \int \cancel{\sin x} \cdot 1 \cdot dx =$

$= \cancel{\sin x} \cdot x - \int \cancel{\sin x} \cdot dx =$ **Atenção!**

$= \cancel{\sin x} \cdot x + \cos x + K.$

onde $K \in \mathbb{R}$

Escolha Cuche

$f'(x) = \cos x$ $f(x) = \sin x$

$g(x) = x$ $g'(x) = 1$